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An image appers too bright?

Suitable Grey level Transformation: Logarithmic (Log) Transforma

* log transformation Comparesses high-intensity pixel val while expanding low-intensity values. This reduces excessive brightness and revels hidden details.

Formula:

$s = c \log(1+r)$

where
- r = input pixel value
- s = output pixel value
- c = scaling constant

Application Process

1) Read the bright images
2) Apply the logarithmic transformation to every pixel
3) Normalize the output intensity values.
4) Display the enhanced image.

Expected output:-

* Excessive brightness is reduced.
* Details in bright regions become visible.

Justification

Since log transformation compresses higher intensity values,

Q) A dark image lacks detail?

AN) Suitable Grey level Transformation: Power law (Gamma) Transformation ($\gamma<1$)

Concept

Gamma transformation adjusts images brightness by changing the relationship between input and output intensities.

Formula:

$s = cr^{\gamma}$

where:

- $0<\gamma<1$ for dark images.

Application process

1) Read the dark image.
2) Select gamma value less than 1
3) Apply the power-law transformation
4) Normalize intensity values.

Expected output

- Dark regions become brighter
- Hidden details are revealed.

Justification

Gamma valuees less than 1 stretch low-intensity pixels more than high-intensity pixles,

3) An image has poor Contrast

Ans Suitable Histogram-Based Technique: Histogram Equalization

Concept

Histogram Equalization redistributes pixel intensity values so that they occupy the full available intensity range.

working process

1) Compute image histogram.
2) Calculate Cumulative distribution function (CDF)
3) Map old intensity values to new intensity values
4) Generate the enhanced image.

Expected outcome:

* Contrast increases
* Fine details become visible
* intensity values spread across the entire range

Justification

Histogram equalization automatically enhances global contrast and is highly effective for images

4) A low-Contrast image requires uniform intensity distribution?

Ans ~~Suitable~~ Histogram Equalization attempts to produce a nearly uniform distribution of grey levels.

Working Process

1) Calculate histogram.

2) Compute cumulative histogram

3) Transform each pixel using the cumulative distribution

4) Generate the enhanced image.

Effect on intensity levels.

* Pixel values are spread uniformly
* Dynamic rang is increased
* Low and high intensities become more distinguishable.

Justification:-

A uniform intensity distribution maximizes image contrast, making Histogram Equalization the most suitable techinique.

5) An image Shows uneven brightness across regions?

Ans

Suitable Enhancement Technique: Adaptive Histogram Equalization (CLAHE)

Concept

Contrast limited Adaptive Histogram Equalization enhances contrast locally instead of globally. The imag is divided into small regions, and histogram equalization is applied to each region independently

Working Process

1) Divide the image into small blocks
2) Compute histogram for each tile.
3) Limit histogram amplification to avoid noise
4) Equalize each tile separately.

Expected outcome

- Corrects non-Uniform illumination
- Improves local contrast
- Preserves image details.
- Avoids over-enhancement and excessive noise.